

Cyclic Redundancy Check (CRC) Controller Module

This chapter describes the cyclic redundancy check (CRC) controller module.

NOTE: This chapter describes a superset implementation of the CRC module that includes features and functionality that require DMA. Since not all devices have DMA capability, consult your device-specific datasheet to determine applicability of these features and functions to your device being used.

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14.1 Overview

The CRC controller is a module that is used to perform CRC (Cyclic Redundancy Check) to verify the integrity of memory system. A signature representing the contents of the memory is obtained when the contents of the memory are read into CRC controller. The responsibility of CRC controller is to calculate the signature for a set of data and then compare the calculated signature value against a pre-determined good signature value. CRC controller supports two channels to perform CRC calculation on multiple memories in parallel and can be used on any memory system.

14.1.1 Features

The CRC controller offers:

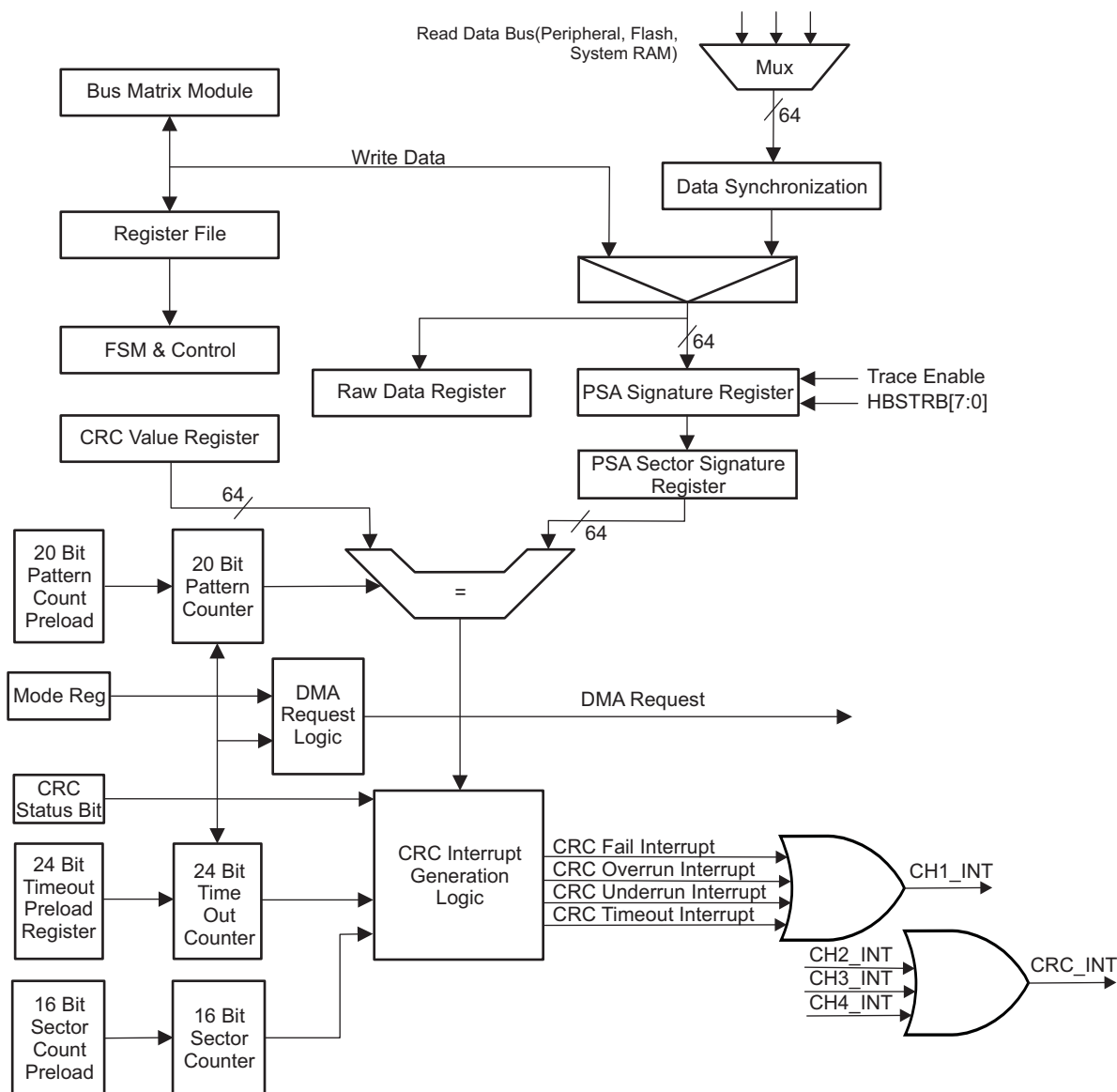
- Two channels to perform background signature verification on any memory sub-system.
- Data compression on 8, 16, 32, and 64 bit data size.
- Maximum-length PSA (Parallel Signature Analysis) register constructed based on 64 bit primitive polynomial.
- Each channel has a CRC Value Register that contains the pre-determined CRC value.
- Use timed base event trigger from timer to initiate DMA data transfer.
- Programmable 20-bit pattern counter per channel to count the number of data patterns for compression.
- Three modes of operation: Auto, Semi-CPU, and Full-CPU.
- For each channel, CRC can be performed either by CRC Controller or by CPU.
- Automatically perform signature verification without CPU intervention in AUTO mode.
- Generate interrupt to CPU in Semi-CPU mode to allow CPU to perform signature verification itself.
- Generate CRC fail interrupt in AUTO mode if signature verification fails.
- Generate Timeout interrupt if CRC is not performed within the time limit.
- Generate DMA request per channel to initiate CRC value transfer.
- Data trace capability on Peripheral Bus Master, Flash and System RAM data buses.

14.1.2 Block Diagram

Figure 14-1 shows a block diagram of the CRC controller.

NOTE: Only Channel 1 can support data trace. See [Section 14.2.11](#).

Figure 14-1. CRC Controller Block Diagram For One Channel



14.2 Module Operation

14.2.1 General Operation

There are two channels in CRC controller and for each channel there is a memory-mapped PSA (Parallel Signature Analysis) Signature Register and a memory-mapped CRC (Cyclic Redundancy Check) Value register. A memory can be organized into multiple sectors with each sector consisting of multiple data patterns. A data pattern can be a 8, 16, 32, or 64 bit data. CRC module performs the signature calculation and compares the signature to a pre-determined value. The PSA Signature Register compresses an incoming data pattern into a signature when it is written. When one sector of data patterns are written into PSA Signature Register, a final signature corresponding to the sector is obtained. CRC Value Register stores the pre-determined signature corresponding to one sector of data patterns. The calculated signature and the pre-determined signature are then compared to each other for signature verification. To minimize CPU's involvement, data patterns transfer can be carried out at the background of CPU using DMA controller. DMA is setup to transfer data from memory from which the contents to be verified to the memory-mapped PSA Signature Register. When DMA transfers data to the memory-mapped PSA Signature Register, a signature is generated. A programmable 20-bit data pattern counter is used for each channel to define the number of data patterns to calculate for each sector. Signature verification can be performed automatically by CRC controller in AUTO mode or by CPU itself in Semi-CPU or Full-CPU mode. In AUTO mode, a self sustained CRC signature calculation can be achieved without any CPU intervention. CRC Controller also provides data trace capability. Channel 1 can perform data trace on CPU data bus. During data trace, channel 1 monitors any data being read on CPU data bus and compresses it. When data trace is enabled for channel 1, all circuits related to DMA request and interrupt generation and counters are disabled.

14.2.2 CRC Modes of Operation

CRC Controller can operate in AUTO, Semi-CPU, and Full-CPU modes.

14.2.2.1 AUTO Mode

In AUTO mode, CRC Controller in conjunction with DMA controller can perform CRC totally without CPU intervention. A sustained transfer of data to both the PSA Signature Register and CRC Value Register are performed in the background of CPU. When a mismatch is detected, an interrupt is generated to CPU. A 16 bit current sector ID register is provided to identify which sector causes a CRC failure.

14.2.2.2 Semi-CPU Mode

In Semi-CPU mode, DMA controller is also utilized to perform data patterns transfer to PSA Signature Register. Instead of performing signature verification automatically, the CRC controller generates an compression complete interrupt to CPU after each sector is compressed. Upon responding to the interrupt the CPU performs the signature verification by reading the calculated signature stored at the PSA Sector Signature Register and compare it to a pre-determined CRC value.

14.2.2.3 Full CPU Mode

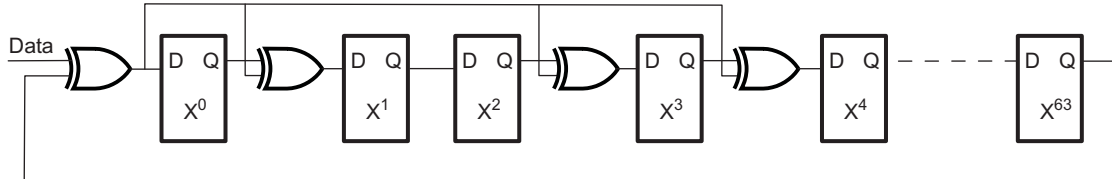
In Full-CPU mode, the CPU does the data patterns transfer and signature verification all by itself. When CPU has enough throughput, it can perform data patterns transfer by reading data from the memory system to the PSA Signature Register. After certain number of data patterns are compressed, the CPU can read from the PSA Signature Register and compare the calculated signature to the pre-determined CRC signature value. In Full-CPU mode, neither interrupt nor DMA request is generated. All counters are also disabled.

14.2.3 PSA Signature Register

The 64-bit PSA Signature Register is based on the primitive polynomial (as in the following equation) to produce the maximum length LFSR (Linear Feedback Shift Register), as shown in Figure 14-2.

$$f(x) = x^{64} + x^4 + x^3 + x + 1 \quad (25)$$

Figure 14-2. LFSR



The serial implementation of LFSF has a limitation that, it requires 'n' clock cycles to calculate the CRC values for an 'n' bit data stream. The idea is to produce the same CRC value operating on a multi-bit data stream, as would occur if the CRC were computed one bit at a time over the whole data stream. The algorithm involves looping to simulate the shifting, and concatenating strings to build the equations after 'n' shift.

The parallel CRC calculation based on the polynomial can be illustrated in the following HDL code:

```
for i in 63 to 0 loop
  NEXT_CRC_VAL(0) := CRC_VAL(63) xor DATA(i);
  for j in 1 to 63 loop
    case j is
      when 1|3|4 =>
        NEXT_CRC_VAL(j) :=
          CRC_VAL(j - 1) xor CRC_VAL(63) xor DATA(i);
      when others =>
        NEXT_CRC_VAL(j) := CRC_VAL(j - 1);
    end case;
  end loop;
  CRC_VAL := NEXT_CRC_VAL;
end loop;
```

-
- NOTE:**
- 1) The inner loop is to calculate the next value of each shift register bit after one cycle
 - 2) The outer loop is to simulate 64 cycles of shifting. The equation for each shift register bit is thus built before it is compressed into the shift register.
 - 3) MSB of the DATA is shifted in first
-

There is one PSA Signature Register per CRC channel. PSA Signature Register can be both read and written. When it is written, it can either compress the data or just capture the data depending on the state of CHx_MODE bits. If CHx_MODE=Data Capture, a seed value can be planted in the PSA Signature Register without compression. Other modes other than Data Capture will result with the data compressed by PSA Signature Register when it is written. Each channel can be planted with different seed value before compression starts. When PSA Signature Register is read, it gives the calculated signature.

CRC Controller should be used in conjunction with the on chip DMA controller to produce optimal system performance. The incoming data pattern to PSA Signature Register is typically initiated by the DMA master. When DMA is properly setup, it would read data from the pre-determined memory system and write them to the memory-mapped PSA Signature Register. Each time PSA Signature Register is written a signature is generated. CPU itself can also perform data transfer by reading from the memory system and perform write operation to PSA Signature Register if CPU has enough throughput to handle data patterns transfer.

After system reset and when AUTO mode is enabled, CRC Controller automatically generates a DMA request to request the pre-determined CRC value corresponding to the first sector of memory to be checked.

In AUTO mode, when one sector of data patterns is compressed, the signature stored at the PSA Signature Register is first copied to the PSA Sector Signature Register and PSA Signature Register is then cleared out to all zeros. An automatic signature verification is then performed by comparing the signature stored at the PSA Sector Signature Register to the CRC Value Register. After the comparison the CRC Controller can generate a DMA request. Upon receiving the DMA request the DMA controller will update the CRC Value Register by transferring the next pre-determined signature value associated with the next sector of memory system. If the signature verification fails then CRC Controller can generate a CRC fail interrupt.

In Full-CPU mode, no DMA request and interrupt are generated at all. The number of data patterns to be compressed is determined by CPU itself. Full-CPU mode is useful when DMA controller is not available to perform background data patterns transfer. The OS can periodically generate a software interrupt to CPU and use CPU to accomplish data transfer and signature verification.

CRC Controller supports doubleword, word, half word and byte access to the PSA Signature Register. During a non-doubleword write access, all unwritten byte lanes are padded with zero's before compression. Note that comparison between PSA Sector Signature Register and CRC Value Register is always in 64 bit because a compressed value is always expressed in 64 bit.

There is a software reset per channel for PSA Signature Register. When set, the PSA Signature Register is reset to all zeros.

PSA Signature Register is reset to zero under the following conditions:

- System reset
- PSA Software reset
- One sector of data patterns are compressed

14.2.4 PSA Sector Signature Register

After one sector of data is compressed, the final resulting signature calculated by PSA Signature Register is transferred to the PSA Sector Signature Register. PSA Signature Register is a read only register. During Semi-CPU mode, the host CPU should read from the PSA Sector Signature Register instead of reading from PSA Signature Register for signature verification to avoid data coherency issue. The PSA Signature Register can be updated with new signature before the host CPU is able to retrieve it.

In Semi-CPU mode, no DMA request is generated. When one sector of data patterns is compressed, CRC controller first generates a compression complete interrupt. Responding to the interrupt, CPU will in the ISR read the PSA Sector Signature Register and compare it to the known good signature or write the signature value to another memory location to build a signature file. In Semi-CPU mode, CPU must perform the signature verification in a manner to prevent any overrun condition. The overrun condition occurs when the compression complete interrupt is generated after one sector of data patterns is compressed and CPU has not read from the PSA Sector Signature Register to perform necessary signature verification before PSA Sector Signature Register is overridden with a new value. An overrun interrupt can be enable to generate when overrun condition occurs. During Semi-CPU mode, the host CPU should read from the PSA Sector Signature Register instead of reading from PSA Signature Register for signature verification to avoid data coherency issue. The PSA Signature Register can be updated with new signature before the host CPU is able to retrieve it.

14.2.5 CRC Value Register

Associated with each channel there is a CRC Value Register. The CRC Value Register stores the pre-determined CRC value. After one sector of data patterns is compressed by PSA Signature Register, CRC Controller can automatically compare the resulting signature stored at the PSA Sector Signature Register with the pre-determined value stored at the CRC Value Register if AUTO mode is enabled. If the signature verification fails, CRC Controller can be enabled to generate an CRC fail interrupt. When the channel is set up for Semi-CPU mode, CRC controller first generates a compression complete interrupt to CPU. Upon servicing the interrupt, CPU will then read the PSA Sector Signature Register and then read the corresponding CRC value stored at another location and compare them. CPU should not read from the CRC Value Register during Semi-CPU or Full-CPU mode because the CRC Value Register is not updated during these two modes.

In AUTO mode, for first sector's signature, DMA request is generated when mode is programmed to AUTO. For subsequent sectors, DMA request is generated after each sector is compressed. Responding to the DMA request, DMA controller reloads the CRC Value Register for the next sector of memory system to be checked.

When CRC Value Register is updated with a new CRC value, an internal flag is set to indicate that CRC Value Register contains the most current value. This flag is cleared when CRC comparison is performed. Each time at the end of the final data pattern compression of a sector, CRC Controller first checks to see if the corresponding CRC Value Register has the most current CRC value stored in it by polling the flag. If the flag is set then the CRC comparison can be performed. If the flag is not set then it means the CRC Value Register contains stale information. A CRC underrun interrupt is generated. When an underrun condition is detected, signature verification is not performed.

CRC Controller supports doubleword, word, half word and byte access to the CRC Value Register. As noted before comparison between PSA Sector Signature Register and CRC Value Register during AUTO mode is carried out in 64 bit.

14.2.6 Raw Data Register

The raw or un-compressed data written to the PSA Signature Register is also saved in the Raw Data Register. This register is read only.

14.2.7 Example DMA Controller Setup

DMA controller needs to be setup properly in either AUTO or Semi-CPU mode as DMA controller is used to transfer data patterns. Hardware or a combination of hardware and software DMA triggering are supported.

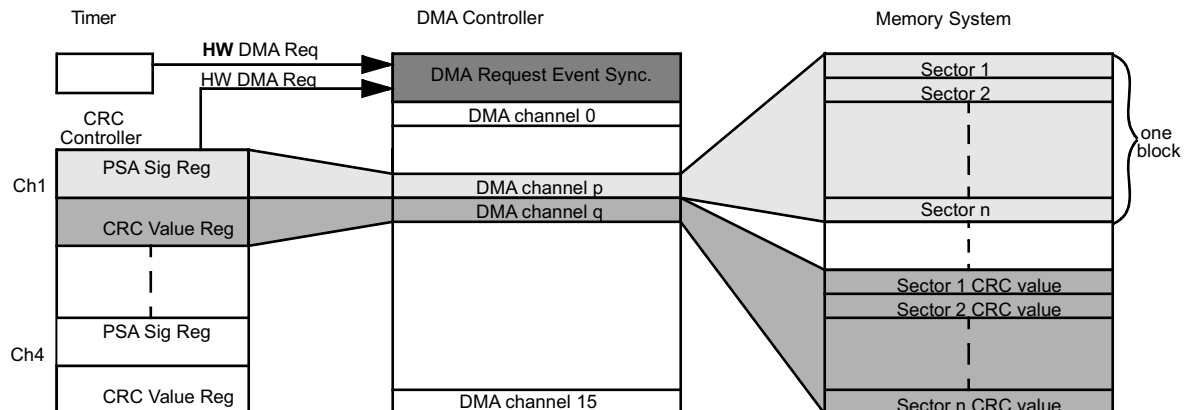
14.2.7.1 AUTO Mode Using Hardware Timer Trigger

There are two DMA channels associated with each CRC channel when in AUTO mode. One DMA channel is setup to transfer data patterns from the source memory to the PSA Signature Register. The second DMA channel is setup to transfer the pre-determined signature to the CRC Value Register. The trigger source for the first DMA channel can be either by hardware or by software. As illustrated in [Figure 14-3](#) a timer can be used to trigger a DMA request to initiate transfer from the source memory system to PSA Signature Register. In AUTO mode, CRC Controller also generates DMA request after one sector of data patterns is compressed to initiate transfer of the next CRC value corresponding to the next sector of memory. Thus a new CRC value is always updated in the CRC Value Register by DMA synchronized to each sector of memory.

A block of memory system is usually divided into many sectors. All sectors are the same size. The sector size is programmed in the CRC_PCOUNT_REGx and the number of sectors in one block is programmed in the CRC_SCOUNT_REGx of the respective channel. CRC_PCOUNT_REGx multiplies CRC_SCOUNT_REGx and multiplies transfer size of each data pattern should give the total block size in number of bytes.

The total size of the memory system to be examined is also programmed in the respective transfer count register inside DMA module. The DMA transfer count register is divided into two parts. They are element count and frame count. Note that an HW DMA request can be programmed to trigger either one frame or one entire block transfer. In [Figure 14-3](#), an HW DMA request from a timer is used as a trigger source to initiate DMA transfer. If all four CRC channels are active in AUTO mode then a total of four DMA requests would be generated by CRC Controller.

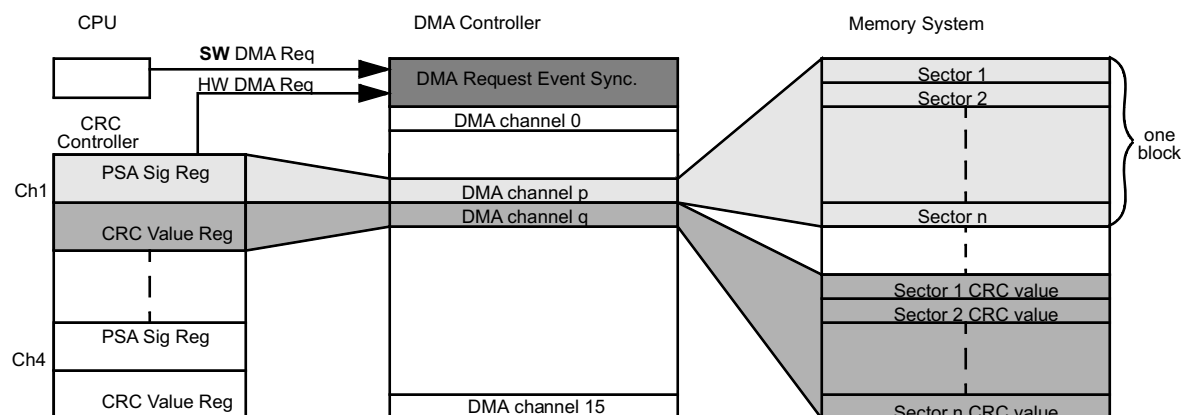
Figure 14-3. AUTO Mode Using Hardware Timer Trigger



14.2.7.2 AUTO Mode Using Software Trigger

The data patterns transfer can also be initiated by software. CPU can generate a software DMA request to activate the DMA channel to transfer data patterns from source memory system to the PSA Signature Register. To generate a software DMA request CPU needs to set the corresponding DMA channel in the DMA software trigger register. Note that just one software DMA request from CPU is enough to complete the entire data patterns transfer for all sectors. See [Figure 14-4](#) for an illustration.

Figure 14-4. AUTO Mode With Software CPU Trigger



14.2.7.3 Semi-CPU Mode Using Hardware Timer Trigger

During semi-CPU mode, no DMA request is generated by CRC controller. Therefore, no DMA channel is allocated to update CRC Value Register. CPU should not read from CRC Value Register in semi-CPU mode as it contains stale value. Note that no signature verification is performed at all during this mode. Similar to AUTO mode, either by hardware or by software DMA request can be used as a trigger for data patterns transfer. Figure 14-5 illustrates the DMA setup using semi-CPU mode with hardware timer trigger.

Figure 14-5. Semi-CPU Mode With Hardware Timer Trigger

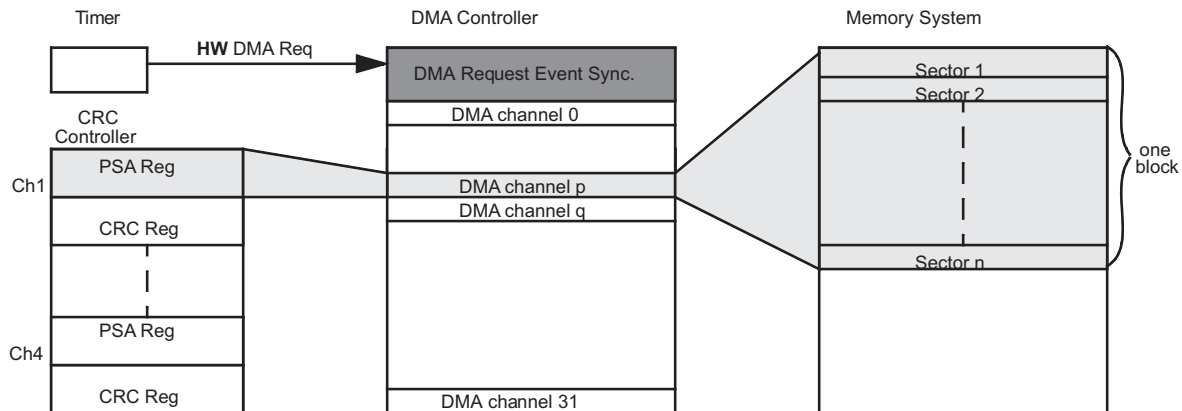


Table 14-1. CRC Modes in Which DMA Request and Counter Logic are Active or Inactive

Mode	DMA Request	Pattern Counter	Sector Counter	Timeout Counter
AUTO	Active	Active	Active	Active
Semi-CPU	Inactive	Active	Active	Active
Full-CPU	Inactive	Inactive	Inactive	Inactive

14.2.8 Pattern Count Register

There is a 20-bit data pattern counter for every CRC channel. The data pattern counter is a down counter and can be pre-loaded with a programmable value stored in the Pattern Count Register. When the data pattern counter reaches zero, a compression complete interrupt is generated in Semi-CPU mode and an automatic signature verification is performed in AUTO mode. In AUTO only, DMA request is generated to trigger the DMA controller to update the CRC Value Register.

NOTE: The data pattern count should be divisible by the total transfer count as programmed in DMA controller. The total transfer count is the product of element count and frame count.

14.2.9 Sector Count Register/Current Sector Register

Each channel contains a 16 bit sector counter. The sector count register stores the number of sectors. Sector counter is a free running counter and is incremented by one each time when one sector of data patterns is compressed. When the signature verification fails, the current value stored in the sector counter is saved into current sector register. If signature verification fails, CPU can read from the current sector register to identify the sector which causes the CRC mismatch. To aid and facilitate the CPU in determining the cause of a CRC failure, it is advisable to use the following equation during CRC and DMA setup:

$$\text{CRC Pattern Count} \times \text{CRC Sector Count} = \text{DMA Element Count} \times \text{DMA Frame Count}$$

The current sector register is frozen from being updated until both the current sector register is read and CRC fail status bit is cleared by CPU. If CPU does not respond to the CRC failure in a timely manner before another sector produces a signature verification failure, the current sector register is not updated with the new sector number. An overrun interrupt is generate instead. If current sector register is already frozen with an erroneous sector and emulation is entered with SUSPEND signal goes to high then the register still remains frozen even it is read.

In Semi-CPU mode, the current sector register is used to indicate the sector for which the compression complete has last happened.

The current sector register is reset when the PSA software reset is enabled.

NOTE: Both data pattern count and sector count registers must be greater than or equal to one for the counters to count. After reset, pattern count and sector count registers default to zero and the associated counters are inactive.

14.2.10 Interrupt

The CRC controller generates several types of interrupts per channel. Associated with each interrupt, there is an interrupt enable bit. No interrupt is generated in Full-CPU mode.

- Compression complete interrupt
- CRC fail interrupt
- Overrun interrupt
- Underrun interrupt
- Timeout interrupt

Table 14-2. Modes in Which Interrupt Condition Can Occur

	AUTO	Semi-CPU	Full-CPU
Compression Complete	no	yes	no
CRC Fail	yes	no	no
Overrun	yes	yes	no
Underrun	yes	no	no
Timeout	yes	yes	no

14.2.10.1 Compression Complete Interrupt

Compression complete interrupt is generated in Semi-CPU mode only. When the data pattern counter reaches zero, the compression complete flag is set and the interrupt is generated.

14.2.10.2 CRC Fail Interrupt

CRC fail interrupt is generated in AUTO mode only. When the signature verification fails, the CRC fail flag is set,. CPU should take action to address the fail condition and clear the CRC fail flag after it resolves the CRC mismatch.

14.2.10.3 Overrun Interrupt

Overrun interrupt is generated in either AUTO or Semi-CPU mode. During AUTO mode, if a CRC fail is detected, then the current sector number is recorded in the current sector register. If CRC fail status bit is not cleared and current sector register is not read by the host CPU before another CRC fail is detected for another sector, then an overrun interrupt is generated. During Semi-CPU mode, when the data pattern counter finishes counting, it generates a compression complete interrupt. At the same time, the signature is copied into the PSA Sector Signature Register. If the host CPU does not read the signature from PSA Sector Signature Register before it is updated again with a new signature value, then an overrun interrupt is generated.

14.2.10.4 Underrun Interrupt

Underrun interrupt only occurs in AUTO mode. The interrupt is generated when the CRC Value Register is not updated with the corresponding signature when the data pattern counter finishes counting. During AUTO mode, CRC Controller generates DMA request to update CRC Value Register in synchronization to the corresponding sector of the memory. Signature verification is also performed if underrun condition is detected. And CRC fail interrupt is generated at the same time as the underrun interrupt.

14.2.10.5 Timeout Interrupt

To ensure that the memory system is examined within a pre-defined time frame and no loss of incoming data there is a 24 bit timeout counter per CRC channel. The 24 bit timeout down counter can be pre-loaded with two different pre-load values, watchdog timeout pre-load value (CRC_WDTPLDx) and block complete timeout pre-load value (CRC_BCTOPLDx). The timeout counter is clocked by a prescaler clock which is permanently running at division 64 of HCLK clock.

First pattern of data must be transferred by the DMA before the timeout counter expires, Watchdog timeout pre-load register (CRC_WDTPLDx) is used as timeout counter. Block complete timeout pre-load register (CRC_BCTOPLDx) is used to check if one complete block of data patterns are compressed within a specific time frame. The timeout counter is first pre-loaded with CRC_WDTPLDx after either AUTO or Semi-CPU mode is selected and starts to down count. If the timeout counter expires before DMA transfers any data pattern to PSA Signature Register then a timeout interrupt is generated. An incoming data pattern before the timeout counter expires will automatically pre-load the timeout counter with CRC_BCTOPLDx the block complete timeout pre-load value.

Block complete timeout pre-load value is used to check if one block of data patterns are compressed within a given time limit. If the timeout counter pre-loaded with CRC_BCTOPLDx value expires before one block of data patterns are compressed a timeout interrupt is generated. When one block (pattern count x sector count) of data patterns are compressed before the counter has expired, the counter is pre-loaded with CRC_WDTPLDx value again. If the timeout counter is pre-loaded with zero then the counter is disable and no timeout interrupt is generated.

In [Figure 14-6](#), a timer generates DMA request every 10ms to trigger one block (pattern count x sector count) transfer. Since we want to make sure that DMA does start to transfer a block every 10 ms we would set the first pre-load value to 10ms in CRC_WDTPLDx. We also want to make sure that one block of data patterns are compressed within 4ms. With such a requirement, we would set the second pre-load value to 4ms in CRC_BCTOPLDx register.

Figure 14-6. Timeout Example 1

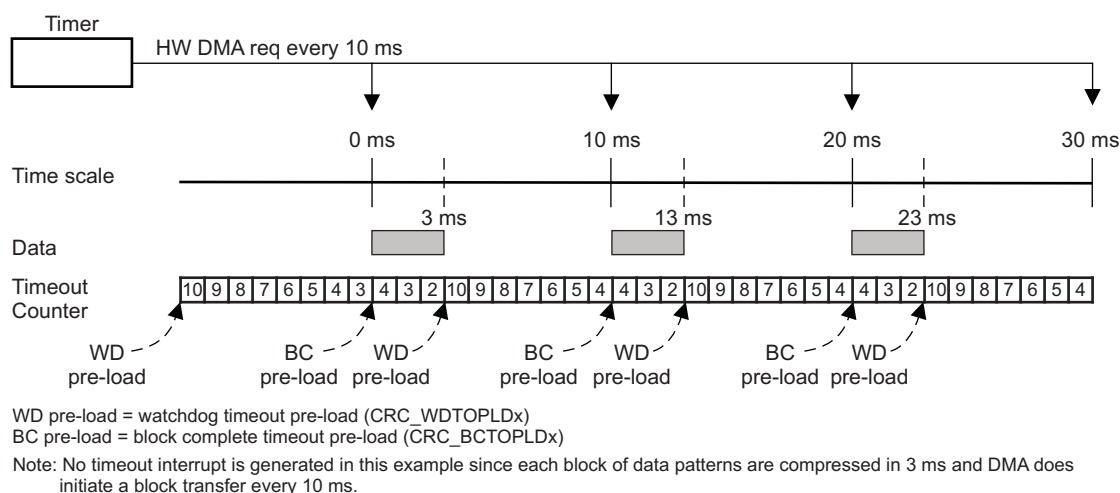
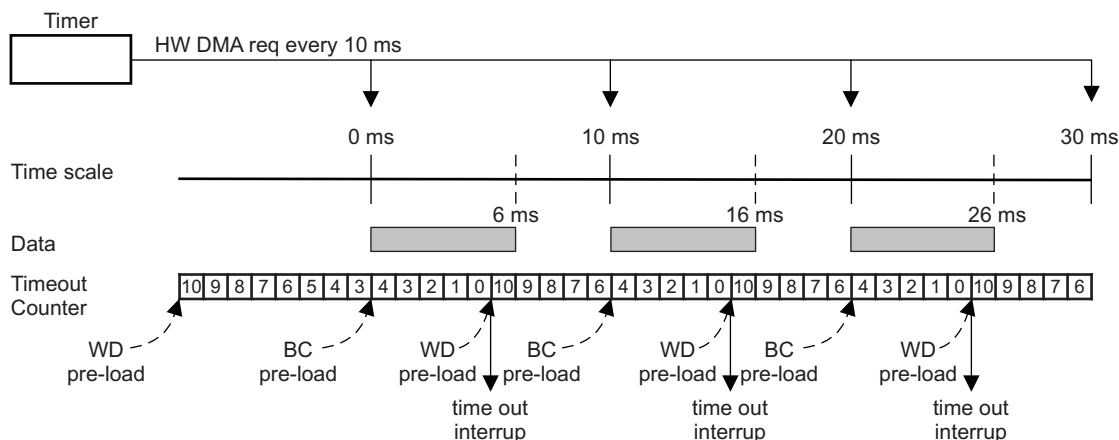
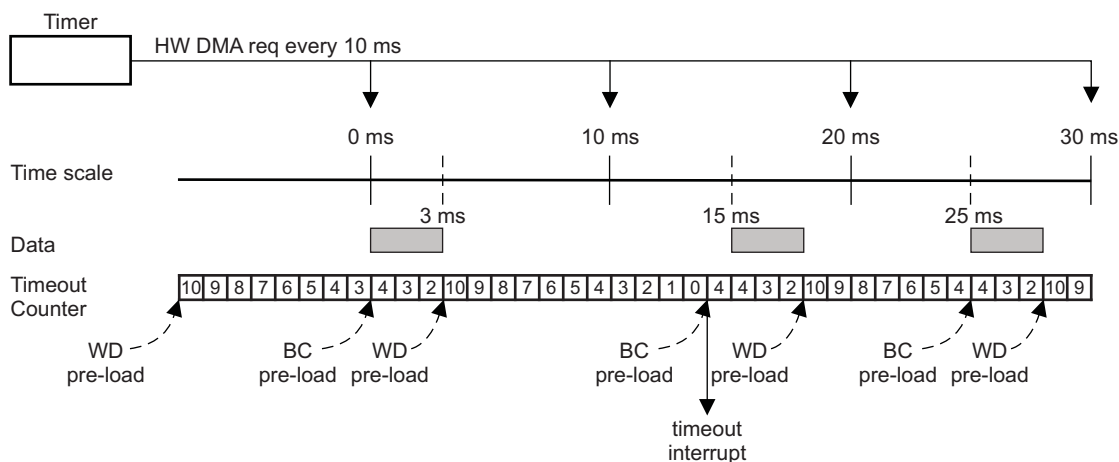


Figure 14-7. Timeout Example 2


WD pre-load = watchdog timeout pre-load (CRC_WDTPLDx)

BC pre-load = block complete timeout pre-load (CRC_BCTPLDx)

Note: Timeout interrupt is generated in this example since each block of data patterns are compressed in 6 ms and this is out of the 4ms time frame.

Figure 14-8. Timeout Example 3


WD pre-load = watchdog timeout pre-load (CRC_WDTPLDx)

BC pre-load = block complete timeout pre-load (CRC_BCTPLDx)

Note: Timeout interrupt is generated in this example since DMA can not transfer the second block of data within 10ms time limit and the reason may be that DMA is set up in fixed priority scheme and DMA is serving other higher priority channels at the time before it can service the timer request.

14.2.10.6 Interrupt Offset Register

CRC Controller only generates one interrupt request to interrupt manager. A interrupt offset register is provided to indicate the source of the pending interrupt with highest priority. [Table 14-3](#) shows the offset interrupt vector address of each interrupt condition in an ascending order of priority.

Table 14-3. Interrupt Offset Mapping

Offset Value	Interrupt Condition
0	Phantom
1h	Ch1 CRC Fail
2h	Ch2 CRC Fail
3h-8h	Reserved
9h	Ch1 Compression Complete
Ah	Ch2 Compression Complete
Bh-10h	Reserved
11h	Ch1 Overrun
12h	Ch2 Overrun
13h-18h	Reserved
19h	Ch1 Underrun
1Ah	Ch2 Underrun
1Bh-20h	Reserved
21h	Ch1 Timeout
22h	Ch2 Timeout
23h-24h	Reserved

14.2.10.7 Error Handling

When an interrupt is generated, host CPU should take appropriate actions to identify the source of error and restart the respective channel in DMA and CRC module. To restart a CRC channel, the user should perform the following steps in the ISR:

1. Write to software reset bit in CRC_CTRL register to reset the respective PSA Signature Register.
2. Reset the CHx_MODE bits to 00 in CRC_CTRL register as Data capture mode.
3. Set the CHx_MODE bits in CRC_CTRL register to desired new mode again.
4. Release software reset.

The host CPU should use byte write to restart each individual channel.

14.2.11 CPU Data Trace

CRC channel 1 can be used to snoop Flash, System RAM and Peripheral Bus Master Data buses. However, at any one point only one bus is snooped. It is possible to disable the snooping of any of the buses by programming CRC_BUS_SEL register. While snooping the data, there is a priority scheme implemented between the buses. Peripheral Bus Master has the highest priority followed by Flash, Even System RAM and Odd System RAM. For each data read by CPU on its data bus the same data is compressed in the PSA Signature Register. A write to PSA Signature Register does not get compressed. Therefore, it is possible to write a seed value into PSA Signature Register before the bus snooping takes place. During data trace mode, all interrupts and DMA request logic are inactive. For non double word read on the data bus, all un-selected byte lanes are padded with zero during compression.

14.2.11.1 Data Capture Mode used in Conjunction with Data Trace

Data capture mode is especially useful when it is used in conjunction when data trace for channel 1 is enabled (CRC_CTRL2.CH1_TRACEEN = 1). The seed value can be planted in PSA Signature Register during data capture mode by writing a seed value into PSA Signature Register. The data trace enable bit is then set to snoop and compress any read transaction on DAHB bus. When trace enable bit is set, CRC_CTRL2.CH1_MODE is automatically reset to 0 (data capture mode). To change from one mode to another mode in the middle of an on-going mode, perform the following steps:

1. Assert software reset for the respective channel.
2. Change from the current active mode to Data Capture mode (CRC_CTRL2.CH1_MODE = 0).
3. Change from Data Capture mode to the new mode.
4. Release software reset.

14.2.12 Power Down Mode

CRC module can be put into power down mode when the power down control bit PWDN is set. The module wakes up when the PWDN bit is cleared. When CRC controller is in power down mode, no data tracing alone will happen. However, if CRC registers are accessed then data trace happens from channel 1.

14.2.13 Emulation

A read access from a register in functional mode can sometimes trigger a certain internal event to follow. For example, reading an interrupt offset register triggers an event to clear the corresponding interrupt status flag. During emulation when SUSPEND signal is high, a read access from any register should only return the register contents to the bus and should not trigger or mask any event as it would have in functional mode. This is to prevent debugger from reading the interrupt offset register during refreshing screen and cause the corresponding interrupt status flag to get cleared. Timeout counters are stopped to generate timeout interrupts in emulation mode. No Peripheral Master bus error should be generated if reading from the unimplemented locations. When channel 1 is placed under data trace, the PSA Signature Register does not compress any data read on CPU data bus when suspend is active.

14.2.14 Peripheral Bus Interface

CRC is a Peripheral slave module. The register interface is similar to other peripheral modules. CRC supports following features:

- Different sizes of burst operation.
- Aligned and unaligned accesses.
- Abort is generated for any illegal address accesses.

14.3 Example

This section illustrates several of the ways in which the CRC Controller can be utilized to perform CRC.

14.3.1 Example: Auto Mode Using Time Based Event Triggering

A large memory area with 2Mbyte (256k doubleword) is to be checked in the background of CPU. CRC is to be performed every 1K byte (128 doubleword). Therefore there should be 2048 pre-recorded CRC values. For illustration purpose, we map channel 1 CRC Value Register to DMA channel 1 and channel 1 PSA Signature Register to DMA channel 2. Assume all DMA transfers are carried out in 64-bit transfer size.

14.3.1.1 DMA Setup

- Set up DMA channel 1 with the starting address from which the pre-determined CRC values are stored. Set up the destination address to the memory-mapped channel 1 CRC Value Register. Put the source address at post increment addressing mode and put the destination address at constant addressing mode. Use **hardware** DMA request for channel 1 to trigger a **frame** transfer.
- Set up DMA channel 2 with the source address from which the contents of memory to be verified. Set up the destination address to the memory-mapped channel 1 PSA Signature Register. Program the element transfer count to 128 and the frame transfer count to 2048. Put the source address at post increment addressing mode and put the destination address at constant address mode. Use **hardware** DMA request for channel 2 to trigger an entire **block** transfer.

14.3.1.2 Timer Setup

The timer can be any general-purpose timer that is capable of generating a time-based DMA request.

- Set up timer to generate DMA request associated with DMA channel 2. For example, an OS can set up the timer to generate a DMA request every 10ms.

14.3.1.3 CRC Setup

- Program the pattern count to 128.
- Program the sector count to 2048.
- For example, we want the entire 2Mbytes to be compressed within 5ms. We can program the block complete timeout pre-load (CRC_BCTOPLDx) value to 15625 ($5 \text{ ms} / (1 \text{ HCLK period} \times 64)$) if CRC is operating at 200 MHz.
- Enable AUTO mode and all interrupts.

After AUTO mode is selected, CRC Controller automatically generates a DMA request on channel 1. Around the same time the timer module also generates a DMA request on DMA channel 2. When the first incoming data pattern arrives at the PSA Signature Register, the CRC Controller will compress it. After some time, the DMA controller would update the CRC Value Register with a pre-determined value matching the calculated signature for the first sector of 128 64 bit data patterns. After one sector of data patterns are compressed, the CRC Controller generate a CRC fail interrupt if signature stored at the PSA Sector Signature Register does not match the CRC Value Register. CRC Controller generates a DMA request on DMA channel 1 when one sector of data patterns are compressed. This routine will continue until the entire 2Mbyte are consumed. If the timeout counter reached zero before the entire 2Mbytes are compressed a timeout interrupt is generated. After 2MBytes are transferred, the DMA can generate an interrupt to CPU. The entire operation will continue again when DMA responds to the DMA request from both the timer and CRC Controller. The CRC is performed totally without any CPU intervention.

14.3.2 Example: Auto Mode Without Using Time Based Triggering

A small but highly secured memory area with 1kbytes is to be checked in the background of CPU. CRC is to be performed every 1Kbytes. Therefore there is only one pre-recorded CRC value. For illustration purpose, we map channel 1 CRC Value Register to DMA channel 1 and channel 1 PSA Signature Register to DMA channel 2. Assume all transfers carried out by DMA are in 64 bit transfer size.

14.3.2.1 DMA Setup

- Set up DMA channel 1 with the source address from which the pre-determined CRC value is stored. Set up the destination address to the memory-mapped channel 1 CRC Value Register. Put the source address at constant addressing mode and put the destination address at constant addressing mode. Use **hardware** DMA request for channel 1.
- Set up DMA channel 2 with the source address from which the memory area to be verified. Set up the destination address to the memory-mapped channel 1 PSA Signature Register. Program the element transfer count to 128 and the frame transfer count to 1. Put the source address at post increment addressing mode and put the destination address at constant address mode. Generate a **software** DMA request on channel 2 after CRC has completed its setup. Enable autoinitiation for DMA channel 2.

14.3.2.2 CRC Setup

- Program the pattern count to 128.
- Program the sector count to 1.
- Leaving the timeout count register with the reset value of zero means no timeout interrupt is generated.
- Enable AUTO mode and all interrupts.

After AUTO mode is selected, the CRC Controller automatically generates a DMA request on channel 1. At the same time the CPU generates a **software** DMA request on DMA channel 2. When the first incoming data pattern arrives at the PSA Signature Register, the CRC Controller will compress it. After some time, the DMA controller would update the CRC Value Register with a pre-determined value matching the calculated signature for the first sector of 128 64 bit data patterns. After one sector of data patterns are compressed, the CRC Controller generates a CRC fail interrupt if signature stored at the PSA Sector Signature Register does not match the CRC Value Register. CRC Controller generates a DMA request on DMA channel 1 again after one sector is compressed. After 1kbytes are transferred, the DMA can generate an interrupt to CPU. Responding to the DMA interrupt CPU can restart the CRC routine by generating a software DMA request onto channel 2 again.

14.3.3 Example: Semi-CPU Mode

If DMA controller is available in a system, the CRC module can also operate in semi-CPU mode. This means that CPU can still make use of the DMA to perform data patterns transfer to CRC controller in the background. The difference between semi-CPU mode and AUTO mode is that CRC controller does not automatically perform the signature verification. CRC controllers generates a compression complete interrupt to CPU when the one sector of data patterns are compressed. CPU needs to perform the signature verification itself.

A memory area with 2Mbyte is to be verified with the help of the CPU. CRC operation is to be performed every 1K byte. Since there are 2Mbyte (256k doublewords) of memory to be check and we want to perform a CRC every 1Kbyte (128 doublewords) and therefore there should be 2048 pre-recorded CRC values. In Semi-CPU mode, the CRC Value Register is not updated and contains indeterminate data.

14.3.3.1 DMA Setup

Set up DMA channel 1 with the source address from which the memory area to be verified are mapped. Set up the destination address to the memory-mapped channel 1 PSA Signature Register. Put the starting address at post increment addressing mode and put the destination address at constant address mode. Use hardware DMA request to trigger an entire block transfer for channel 1. Disable autoinitiation for DMA channel 1.

14.3.3.2 Timer Setup

The timer can be any general-purpose timer that is capable of generating a time-based DMA request.

Set up timer to generate DMA request associated with DMA channel 1. For example, an OS can set up the timer to generate a DMA request every 10ms.

14.3.3.3 CRC Setup

- Program the pattern count to 128.
- Program the sector count to 2048.
- For example, we want the entire 2Mbytes to be compressed within 5ms. We can program the block complete timeout pre-load value to 15625 ($5 \text{ ms} / (1 \text{ HCLK period} \times 64)$) if CRC is operating at 200 MHz.
- Enable Semi-CPU mode and enable all interrupts.

The timer module first generates a DMA request on DMA channel 1 when it is enabled. When the first incoming data pattern arrives at the PSA Signature Register, the CRC controller will compress it. After one sector of data patterns are compressed, the CRC controller generate a compression complete interrupt. Upon responding to the interrupt the CPU would read from the PSA Sector Signature Register. It is up to the CPU on how to deal with the PSA value just read. It can compare it to a known signature value or it can write it to another memory location to build a signature file or even transfer the signature out of the device via SCI or SPI. This routine will continue until the entire 2Mbyte are consumed. The latency of the interrupt response from CPU can cause overrun condition. If CPU does not read from PSA Sector Signature Register before the PSA value is overridden with the signature of the next sector of memory, an overrun interrupt will be generated by CRC controller.

14.3.4 Example: Full-CPU Mode

In a system without the availability of DMA controller, the CRC routine can be operated by CPU provided the CPU has enough throughput. CPU needs to read from the memory area from which CRC is to be performed.

A memory area with 2Mbyte is to be checked with the help of the CPU. CRC verification is to be performed every 1K byte. In CPU mode, the CRC Value Register is not updated and contains indeterminate data.

14.3.4.1 CRC Setup

- All control registers can be left in their reset state. Only enable Full-CPU mode.

CPU itself reads from the memory and write the data to the PSA Signature Register inside CRC Controller. When the first incoming data pattern arrives at the PSA Signature Register, the CRC Controller will compress it. After **2MBytes** data patterns are compressed, CPU can read from the PSA Signature Register. It is up to the CPU on how to deal with the PSA signature value just read. It can compare it to a known signature value stored at another memory location.

14.4 CRC Control Registers

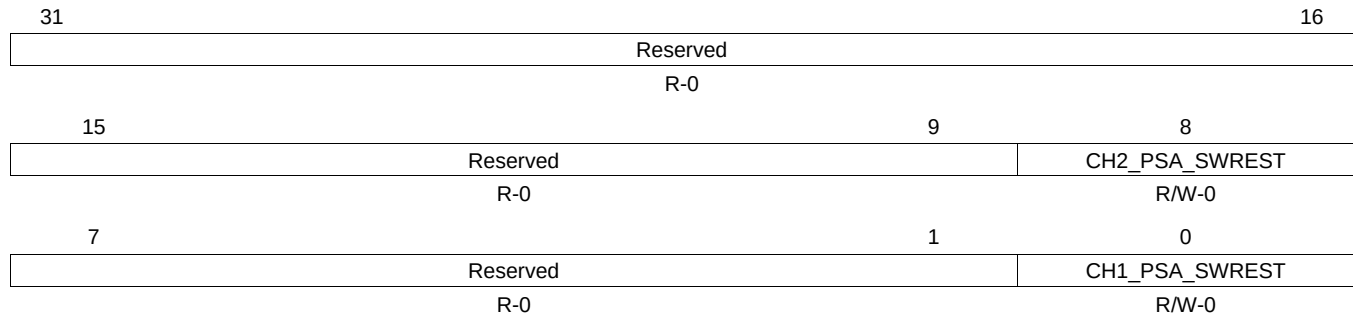
All registers are in word boundary. 64-, 32-, 16-, and 8-bit write accesses are supported to all registers. The base address for the control registers is FE00 0000h.

Table 14-4. CRC Control Registers

Offset	Acronym	Register Description	Section
0h	CRC_CTRL0	CRC Global Control Register	Section 14.4.1
8h	CRC_CTRL1	CRC Global Control Register 1	Section 14.4.2
10h	CRC_CTRL2	CRC Global Control Register 2	Section 14.4.3
18h	CRC_INTS	CRC Interrupt Enable Set Register	Section 14.4.4
20h	CRC_INTR	CRC Interrupt Enable Reset Register	Section 14.4.5
28h	CRC_STATUS	CRC Interrupt Status Register	Section 14.4.6
30h	CRC_INT_OFFSETS_ET_REG	CRC Interrupt Offset Register	Section 14.4.7
38h	CRC_BUSY	CRC Busy Register	Section 14.4.8
40h	CRC_PCOUNT_REG1	CRC Channel 1 Pattern Counter Preload Register	Section 14.4.9
44h	CRC_SCOUNT_REG1	CRC Channel 1 Sector Counter Preload Register	Section 14.4.10
48h	CRC_CURSEC_REG1	CRC Channel 1 Current Sector Register	Section 14.4.11
4Ch	CRC_WDTPD1	CRC Channel 1 Watchdog Timeout Preload Register	Section 14.4.12
50h	CRC_BCTOPD1	CRC Channel 1 Block Complete Timeout Preload Register	Section 14.4.13
60h	PSA_SIGREGL1	Channel 1 PSA Signature Low Register	Section 14.4.14
64h	PSA_SIGREGH1	Channel 1 PSA Signature High Register	Section 14.4.15
68h	CRC_REGL1	Channel 1 CRC Value Low Register	Section 14.4.16
6Ch	CRC_REGH1	Channel 1 CRC Value High Register	Section 14.4.17
70h	PSA_SECSIGREGL1	Channel 1 PSA Sector Signature Low Register	Section 14.4.18
74h	PSA_SECSIGREGH1	Channel 1 PSA Sector Signature High Register	Section 14.4.19
78h	RAW_DATAAREGL1	Channel 1 Raw Data Low Register	Section 14.4.20
7Ch	RAW_DATAAREGH1	Channel 1 Raw Data High Register	Section 14.4.21
80h	CRC_PCOUNT_REG2	CRC Channel 2 Pattern Counter Preload Register	Section 14.4.22
84h	CRC_SCOUNT_REG2	CRC Channel 2 Sector Counter Preload Register	Section 14.4.23
88h	CRC_CURSEC_REG2	CRC Current Sector Register 2	Section 14.4.24
8Ch	CRC_WDTPD2	CRC Channel 2 Watchdog Timeout Preload Register A	Section 14.4.25
90h	CRC_BCTOPD2	CRC Channel 2 Block Complete Timeout Preload Register B	Section 14.4.26
A0h	PSA_SIGREGL2	Channel 2 PSA Signature Low Register	Section 14.4.27
A4h	PSA_SIGREGH2	Channel 2 PSA Signature High Register	Section 14.4.28
A8h	CRC_REGL2	Channel 2 CRC Value Low Register	Section 14.4.29
ACH	CRC_REGH2	Channel 2 CRC Value High Register	Section 14.4.30
B0h	PSA_SECSIGREGL2	Channel 2 PSA Sector Signature Low Register	Section 14.4.31
B4h	PSA_SECSIGREGH2	Channel 2 PSA Sector Signature High Register	Section 14.4.32
B8h	RAW_DATAAREGL2	Channel 2 Raw Data Low Register	Section 14.4.33
BCh	RAW_DATAAREGH2	Channel 2 Raw Data High Register	Section 14.4.34
140h	CRC_BUS_SEL	Data Bus Selection Register	Section 14.4.35

14.4.1 CRC Global Control Register 0 (CRC_CTRL0)

Figure 14-9. CRC Global Control Register 0 (CRC_CTRL0) [offset = 00h]



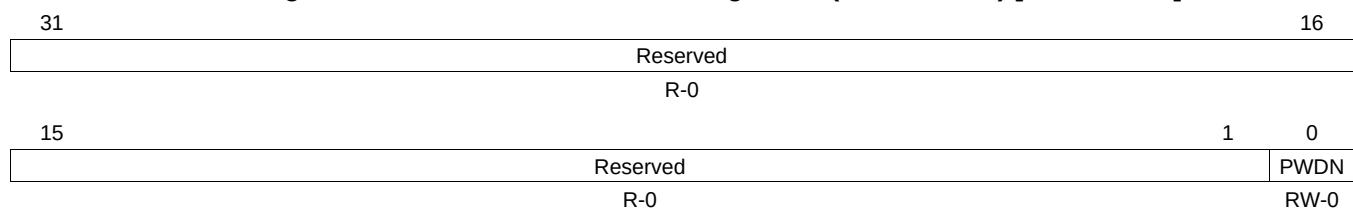
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-5. CRC Global Control Register 0 (CRC_CTRL0) Field Descriptions

Bit	Field	Value	Description
31-9	Reserved	0	Reads return 0. Writes have no effect.
8	CH2_PSA_SWREST	0 1	Channel 2 PSA Software Reset. When set, the PSA Signature Register is reset to all zero. Software reset does not reset software reset bit itself. Therefore, CPU is required to clear this bit by writing a 0. PSA Signature Register is not reset. PSA Signature Register is reset.
7-1	Reserved	0	Reads return 0. Writes have no effect.
0	CH1_PSA_SWREST	0 1	Channel 1 PSA Software Reset. When set, the PSA Signature Register is reset to all zero. Software reset does not reset software reset bit itself. Therefore, CPU is required to clear this bit by writing a 0. PSA Signature Register is not reset. PSA Signature Register is reset.

14.4.2 CRC Global Control Register 1 (CRC_CTRL1)

Figure 14-10. CRC Global Control Register 1 (CRC_CTRL1) [offset = 08h]



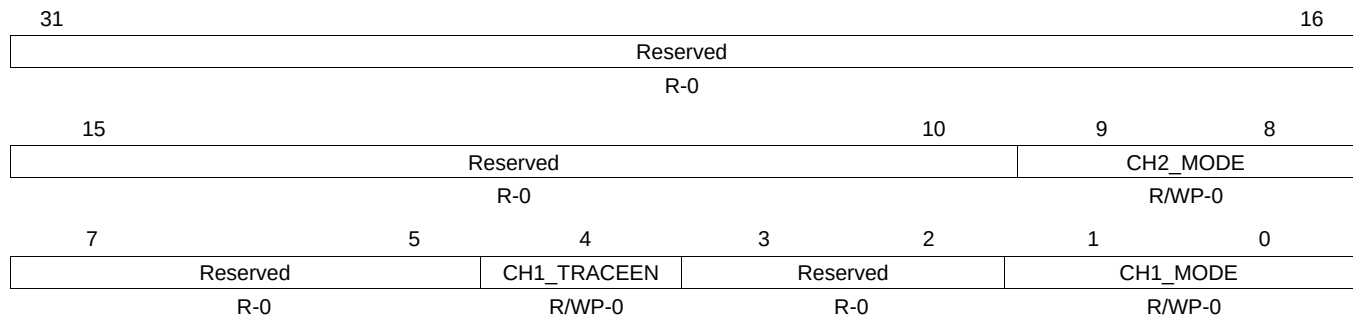
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-6. CRC Global Control Register 1 (CRC_CTRL1) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	PWDN	0 1	Power Down. When set, CRC module is put in power-down mode CRC is not in power-down mode. CRC is in power-down mode.

14.4.3 CRC Global Control Register 2 (CRC_CTRL2)

Figure 14-11. CRC Global Control Register 2 (CRC_CTRL2) [offset = 10h]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 14-7. CRC Global Control Register 2 (CRC_CTRL2) Field Descriptions

Bit	Field	Value	Description
31-10	Reserved	0	Reads return 0. Writes have no effect.
9-8	CH2_MODE	0 1h 2h 3h	Channel 2 Mode Selection 0 Data Capture mode. In this mode, the PSA Signature Register does not compress data when it is written. Any data written to PSA Signature Register is simply captured by PSA Signature Register without any compression. This mode can be used to plant seed value into the PSA register. 1h AUTO Mode 2h Semi-CPU Mode 3h Full-CPU Mode
7-5	Reserved	0	Reads return 0. Writes have no effect.
4	CH1_TRACEEN	0 1	Channel 1 Data Trace Enable. When set, the channel is put into data trace mode. The channel snoops on the CPU Peripheral Bus Master, Flash, System RAM buses for any read transaction. Any read data on these buses is compressed by the PSA Signature Register. When suspend is on, the PSA Signature Register does not compress any read data on these buses. When trace enable bit is set, the CH1_MODE bit is automatically reset to 0 (Data Capture mode). 0 Data Trace is disabled. 1 Data Trace is enabled.
3-2	Reserved	0	Reads return 0. Writes have no effect.
1-0	CH1_MODE	0 1h 2h 3h	Channel 1 Mode Selection 0 Data Capture mode. In this mode, the PSA Signature Register does not compress data when it is written. Any data written to PSA Signature Register is simply captured by PSA Signature Register without any compression. This mode can be used to plant seed value into the PSA register. 1h AUTO mode 2h Semi-CPU mode 3h Full-CPU mode

14.4.4 CRC Interrupt Enable Set Register (CRC_INTS)

Figure 14-12. CRC Interrupt Enable Set Register (CRC_INTS) [offset = 18h]

31											16
Reserved											
R-0											
15	13			12	11	10	9		8		
Reserved				CH2_ TIMEOUTENS	CH2_ UNDERENS	CH2_ OVERENS	CH2_ CRCFAILENS		CH2_ CCITENS		
R-0				R/WP-0	R/WP-0	R/WP-0	R/WP-0		R/WP-0		
7	5			4	3	2	1		0		
Reserved				CH1_ TIMEOUTENS	CH1_ UNDERENS	CH1_ OVERENS	CH1_ CRCFAILENS		CH1_ CCITENS		
R-0				R/WP-0	R/WP-0	R/WP-0	R/WP-0		R/WP-0		

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 14-8. CRC Interrupt Enable Set Register (CRC_INTS) Field Descriptions

Bit	Field	Value	Description
31-13	Reserved	0	Reads return 0. Writes have no effect.
12	CH2_TIMEOUTENS	0 1	Channel 2 Timeout Interrupt Enable Bit. User and Privileged mode (read): 0 Timeout Interrupt is disabled. 1 Timeout Interrupt is enabled.
		0 1	Privileged mode (write): 0 Has no effect. 1 Timeout Interrupt is enabled.
11	CH2_UNDERENS	0 1	Channel 2 Underrun Interrupt Enable Bit. User and Privileged mode (read): 0 Underrun Interrupt is disabled. 1 Underrun Interrupt is enabled.
		0 1	Privileged mode (write): 0 Has no effect. 1 Underrun Interrupt is enabled.
10	CH2_OVERENS	0 1	Channel 2 Overrun Interrupt Enable Bit. User and Privileged mode (read): 0 Overrun Interrupt is disabled. 1 Overrun Interrupt is enabled.
		0 1	Privileged mode (write): 0 Has no effect. 1 Overrun Interrupt is enabled.
9	CH2_CRCFAILENS	0 1	Channel 2 CRC Fail Interrupt Enable Bit. User and Privileged mode (read): 0 CRC Fail Interrupt is disabled. 1 CRC Fail Interrupt is enabled.
		0 1	Privileged mode (write): 0 Has no effect. 1 CRC Fail Interrupt is enabled.

Table 14-8. CRC Interrupt Enable Set Register (CRC_INTS) Field Descriptions (continued)

Bit	Field	Value	Description
8	CH2_CCITENS		Channel 2 Compression Complete Interrupt Enable Bit.
		0	User and Privileged mode (read): Compression Complete Interrupt is disabled.
		1	Compression Complete Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	Compression Complete Interrupt is enabled.
7-5	Reserved	0	Reads return 0. Writes have no effect.
4	CH1_TIMEOUTENS		Channel 1 Timeout Interrupt Enable Bit.
		0	User and Privileged mode (read): Timeout Interrupt is disabled.
		1	Timeout Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	Timeout Interrupt is enabled.
3	CH1_UNDERENS		Channel 1 Underrun Interrupt Enable Bit.
		0	User and Privileged mode (read): Underrun Interrupt is disabled.
		1	Underrun Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	Underrun Interrupt is enabled.
2	CH1_OVERENS		Channel 1 Overrun Interrupt Enable Bit.
		0	User and Privileged mode (read): Overrun Interrupt is disabled.
		1	Overrun Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	Overrun Interrupt is enabled.
1	CH1_CRCFAILENS		Channel 1 CRC Fail Interrupt Enable Bit.
		0	User and Privileged mode (read): CRC Fail Interrupt is disabled.
		1	CRC Fail Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	CRC Fail Interrupt is enabled.
0	CH1_CCITENS		Channel 1 Compression Complete Interrupt Enable Bit.
		0	User and Privileged mode (read): Compression Complete Interrupt is disabled.
		1	Compression Complete Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	Compression Complete Interrupt is enabled.

14.4.5 CRC Interrupt Enable Reset Register (CRC_INTR)

Figure 14-13. CRC Interrupt Enable Reset Register (CRC_INTR) [offset = 20h]

31	Reserved														16
R-0															
15	13				12	11		10		9		8			
Reserved					CH2_ TIMEOUTENR	CH2_ UNDERENR	CH2_ OVERENR	CH2_ CRCFAILENR	CH2_ CCITENR						
R-0					R/WP-0		R/WP-0		R/WP-0		R/WP-0		R/WP-0		
7	5				4	3		2		1		0			
Reserved					CH1_ TIMEOUTENR	CH1_ UNDERENR	CH1_ OVERENR	CH1_ CRCFAILENR	CH1_ CCITENR						
R-0					R/WP-0		R/WP-0		R/WP-0		R/WP-0		R/WP-0		

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 14-9. CRC Interrupt Enable Reset Register (CRC_INTR) Field Descriptions

Bit	Field	Value	Description
31-13	Reserved	0	Reads return 0. Writes have no effect.
12	CH2_TIMEOUTENR	0 1	Channel 2 Timeout Interrupt Enable Reset Bit. User and Privileged mode (read): 0 Timeout Interrupt is disabled. 1 Timeout Interrupt is enabled.
		0 1	Privileged mode (write): 0 Has no effect. 1 Timeout Interrupt is disabled.
11	CH2_UNDERENR	0 1	Channel 2 Underrun Interrupt Enable Reset Bit. User and Privileged mode (read): 0 Underrun Interrupt is disabled. 1 Underrun Interrupt is enabled.
		0 1	Privileged mode (write): 0 Has no effect. 1 Underrun Interrupt is disabled.
10	CH2_OVERENR	0 1	Channel 2 Overrun Interrupt Enable Reset Bit. User and Privileged mode (read): 0 Overrun Interrupt is disabled 1 Overrun Interrupt is enabled
		0 1	Privileged mode (write): 0 Has no effect 1 Overrun Interrupt disable
9	CH2_CRCFAILENR	0 1	Channel 2 CRC Fail Interrupt Enable Reset Bit. User and Privileged mode (read): 0 CRC Fail Interrupt is disabled. 1 CRC Fail Interrupt is enabled.
		0 1	Privileged mode (write): 0 Has no effect. 1 CRC Fail Interrupt is disabled.

Table 14-9. CRC Interrupt Enable Reset Register (CRC_INTR) Field Descriptions (continued)

Bit	Field	Value	Description
8	CH2_CCITENR		Channel 2 Compression Complete Interrupt Enable Reset Bit.
		0	User and Privileged mode (read): Compression Complete Interrupt is disabled.
		1	Compression Complete Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	Compression Complete Interrupt is disabled.
7-5	Reserved	0	Reads return 0. Writes have no effect.
4	CH1_TIMEOUTENR		Channel 1 Timeout Interrupt Enable Reset Bit.
		0	User and Privileged mode (read): Timeout Interrupt is disabled.
		1	Timeout Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	Timeout Interrupt is disabled.
3	CH1_UNDERENR		Channel 1 Underrun Interrupt Enable Reset Bit.
		0	User and Privileged mode (read): Underrun Interrupt is disabled.
		1	Underrun Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	Underrun Interrupt is disabled.
2	CH1_OVERENR		Channel 1 Overrun Interrupt Enable Reset Bit.
		0	User and Privileged mode (read): Overrun Interrupt is disabled.
		1	Overrun Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	Overrun Interrupt is disabled.
1	CH1_CRCFAILENR		Channel 1 CRC Fail Interrupt Enable Reset Bit.
		0	User and Privileged mode (read): CRC Fail Interrupt is disabled.
		1	CRC Fail Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	CRC Fail Interrupt is disabled.
0	CH1_CCITENR		Channel 1 Compression Complete Interrupt Enable Reset Bit.
		0	User and Privileged mode (read): Compression Complete Interrupt is disabled.
		1	Compression Complete Interrupt is enabled.
			Privileged mode (write):
		0	Has no effect.
		1	Compression Complete Interrupt is disabled.

14.4.6 CRC Interrupt Status Register (CRC_STATUS)

Figure 14-14. CRC Interrupt Status Register (CRC_STATUS) [offset = 28h]

31																	16			
Reserved																				
R-0																				
15					13			12			11			10			9			8
Reserved					CH2_TIMEOUT			CH2_UNDER			CH2_OVER			CH2_CRCFAIL			CH2_CCIT			
R-0					R/W1CP-0			R/W1CP-0			R/W1CP-0			R/W1CP-0			R/W1CP-0			
7					5			4			3			2			1			0
Reserved					CH1_TIMEOUT			CH1_UNDER			CH1_OVER			CH1_CRCFAIL			CH1_CCIT			
R-0					R/W1CP-0			R/W1CP-0			R/W1CP-0			R/W1CP-0			R/W1CP-0			

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 to clear in privilege mode only; -n = value after reset

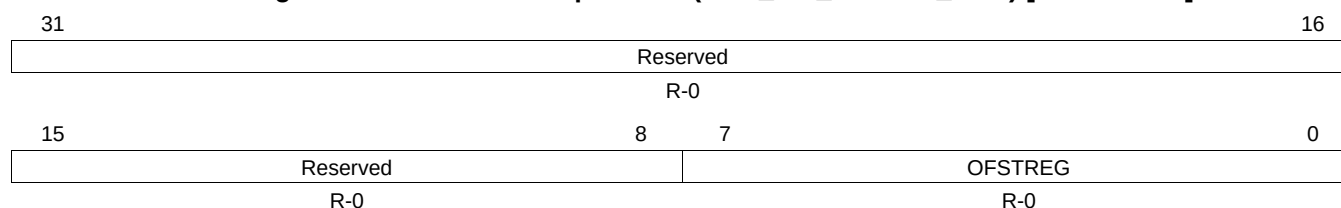
Table 14-10. CRC Interrupt Status Register (CRC_STATUS) Field Descriptions

Bit	Field	Value	Description
31-13	Reserved	0	Reads return 0. Writes have no effect.
12	CH2_TIMEOUT	0 1	Channel 2 CRC Timeout Interrupt Status Flag. This bit is set in both AUTO and Semi-CPU mode. User and Privileged mode (read): 0 No timeout interrupt is active. 1 Timeout interrupt is active.
		0 1	Privileged mode (write): 0 Has no effect. 1 Bit is cleared.
11	CH2_UNDER	0 1	Channel 2 CRC Underrun Interrupt Status Flag. This bit is set in AUTO mode only. User and Privileged mode (read): 0 No Underrun Interrupt is active. 1 Underrun Interrupt is active.
		0 1	Privileged mode (write): 0 Has no effect. 1 Bit is cleared.
10	CH2_OVER	0 1	Channel 2 CRC Overrun Interrupt Status Flag. This bit is set in either AUTO or Semi-CPU mode. User and Privileged mode (read): 0 No Overrun Interrupt is active. 1 Overrun Interrupt is active.
		0 1	Privileged mode (write): 0 Has no effect. 1 Bit is cleared.
9	CH2_CRCFAIL	0 1	Channel 2 CRC Compare Fail Interrupt Status Flag. This bit is set in AUTO mode only. User and Privileged mode (read): 0 No CRC Fail Interrupt is active. 1 CRC Fail Interrupt is active.
		0 1	Privileged mode (write): 0 Has no effect. 1 Bit is cleared.

Table 14-10. CRC Interrupt Status Register (CRC_STATUS) Field Descriptions (continued)

Bit	Field	Value	Description
8	CH2_CCIT		Channel 2 CRC Pattern Compression Complete Interrupt Status Flag. This bit is only set in Semi-CPU mode.
		0	User and Privileged mode (read): No Compression Complete Interrupt is active.
		1	Compression Complete Interrupt is active.
		0	Privileged mode (write): Has no effect.
		1	Bit is cleared.
7-5	Reserved	0	Reads return 0. Writes have no effect.
4	CH1_TIMEOUT		Channel 1 CRC Timeout Interrupt Status Flag.
		0	User and Privileged mode (read): No timeout interrupt is active.
		1	Timeout interrupt is active.
		0	Privileged mode (write): Has no effect.
		1	Bit is cleared.
3	CH1_UNDER		Channel 1 Underrun Interrupt Status Flag.
		0	User and Privileged mode (read): No Underrun Interrupt is active.
		1	Underrun Interrupt is active.
		0	Privileged mode (write): Has no effect.
		1	Bit is cleared.
2	CH1_OVER		Channel 1 Overrun Interrupt Status Flag.
		0	User and Privileged mode (read): No Overrun Interrupt is active.
		1	Overrun Interrupt is active.
		0	Privileged mode (write): Has no effect.
		1	Bit is cleared.
1	CH1_CRCFAIL		Channel 1 CRC Compare Fail Interrupt Status Flag.
		0	User and Privileged mode (read): No CRC Fail Interrupt is active.
		1	CRC Fail Interrupt is active.
		0	Privileged mode (write): Has no effect.
		1	Bit is cleared.
0	CH1_CCIT		Channel 1 CRC Pattern Compression Complete Interrupt Status Flag.
		0	User and Privileged mode (read): No Compression Complete Interrupt is active.
		1	Compression Complete Interrupt is active.
		0	Privileged mode (write): Has no effect.
		1	Bit is cleared.

14.4.7 CRC Interrupt Offset (CRC_INT_OFFSET_REG)

Figure 14-15. CRC Interrupt Offset (CRC_INT_OFFSET_REG) [offset = 30h]


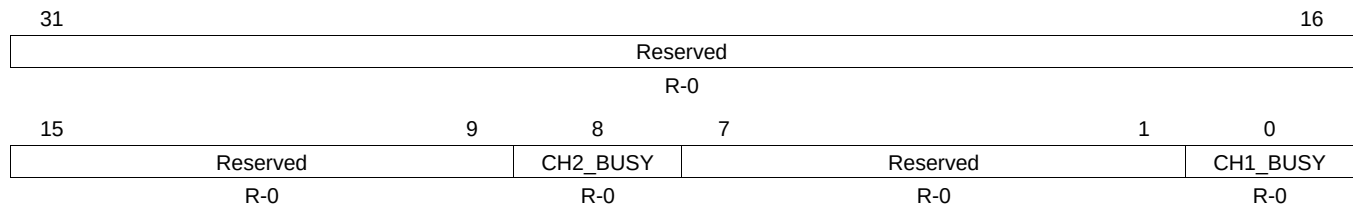
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-11. CRC Interrupt Offset (CRC_INT_OFFSET_REG) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	OFSTREG		CRC Interrupt Offset. This register indicates the highest priority pending interrupt vector address. Reading the offset register automatically clears the respective interrupt flag.
		0	Phantom
		1h	Ch1 CRC Fail
		2h	Ch2 CRC Fail
		3h-8h	Reserved
		9h	Ch1 Compression Complete
		Ah	Ch2 Compression Complete
		Bh-10h	Reserved
		11h	Ch1 Overrun
		12h	Ch2 Overrun
		13h-18h	Reserved
		19h	Ch1 Underrun
		1Ah	Ch2 Underrun
		1Bh-20h	Reserved
		21h	Ch1 Timeout
		22h	Ch2 Timeout
		23h-FFh	Reserved

14.4.8 CRC Busy Register (CRC_BUSY)

Figure 14-16. CRC Busy Register (CRC_BUSY) [offset = 38h]



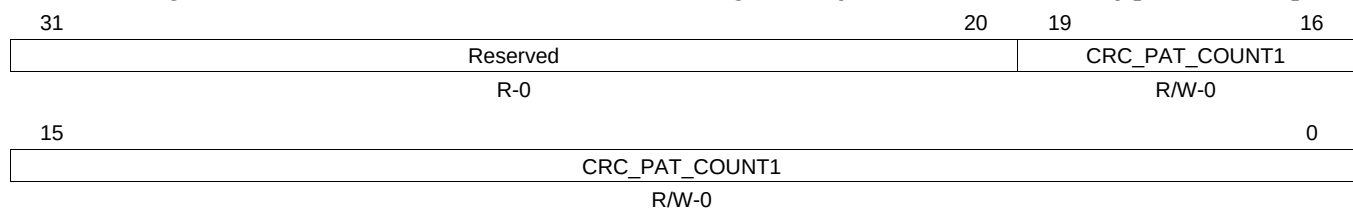
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-12. CRC Busy Register (CRC_BUSY) Field Descriptions

Bit	Field	Value	Description
31-9	Reserved	0	Reads return 0. Writes have no effect.
8	CH2_BUSY		CH2_BUSY. During AUTO or Semi-CPU mode, the busy flag is set when the first data pattern of the block is compressed and remains set until the last data pattern of the block is compressed. The flag is cleared when the last data pattern of the block is compressed.
7-1	Reserved	0	Reads return 0. Writes have no effect.
0	CH1_BUSY		CH1_BUSY. During AUTO or Semi-CPU mode, the busy flag is set when the first data pattern of the block is compressed and remains set until the last data pattern of the block is compressed. The flag is cleared when the last data pattern of the block is compressed.

14.4.9 CRC Pattern Counter Preload Register 1 (CRC_PCOUNT_REG1)

Figure 14-17. CRC Pattern Counter Preload Register 1 (CRC_PCOUNT_REG1) [offset = 40h]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-13. CRC Pattern Counter Preload Register 1 (CRC_PCOUNT_REG1) Field Descriptions

Bit	Field	Value	Description
31-20	Reserved	0	Reads return 0. Writes have no effect.
19-0	CRC_PAT_COUNT1		Channel 1 Pattern Counter Preload Register. This register contains the number of data patterns in one sector to be compressed before a CRC is performed.

14.4.10 CRC Sector Counter Preload Register 1 (CRC_SCOUNT_REG1)

Figure 14-18. CRC Sector Counter Preload Register 1 (CRC_SCOUNT_REG1) [offset = 44h]

31	16
Reserved	
R-0	
15	0
CRC_SEC_COUNT1	
R/W-0	

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-14. CRC Sector Counter Preload Register 1 (CRC_SCOUNT_REG1) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	CRC_SEC_COUNT1		Channel 1 Sector Counter Preload Register. This register contains the number of sectors in one block of memory.

14.4.11 CRC Current Sector Register 1 (CRC_CURSEC_REG1)

Figure 14-19. CRC Current Sector Preload Register 1 (CRC_CURSEC_REG1) [offset = 48h]

31	16
Reserved	
R-0	
15	0
CRC_CURSEC1	
R-0	

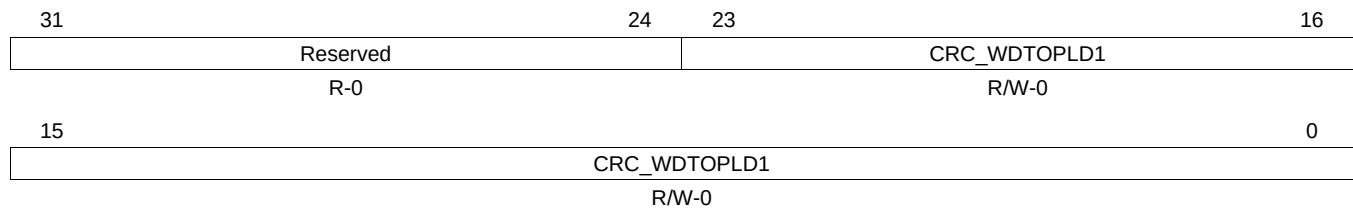
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-15. CRC Current Sector Register 1 (CRC_CURSEC_REG1) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	CRC_CURSEC1		Channel 1 Current Sector ID Register. In AUTO mode, this register contains the current sector number of which the signature verification fails. The sector counter is a free running up counter. When a sector fails, the erroneous sector number is logged into current sector ID register and the CRC fail interrupt is generated. The sector ID register is frozen until it is read and the CRC fail status bit is cleared by CPU. While it is frozen, it does not capture another erroneous sector number. When this condition happens, an overrun interrupt is generated instead. Once the register is read and the CRC fail interrupt flag is cleared it can capture new erroneous sector number. In Semi-CPU mode, this register is used to indicate the sector number for which the compression complete has last happened.

14.4.12 CRC Channel 1 Watchdog Timeout Preload Register A (CRC_WDTPLD1)

Figure 14-20. CRC Channel 1 Watchdog Timeout Preload Register A (CRC_WDTPLD1)
[offset = 4Ch]



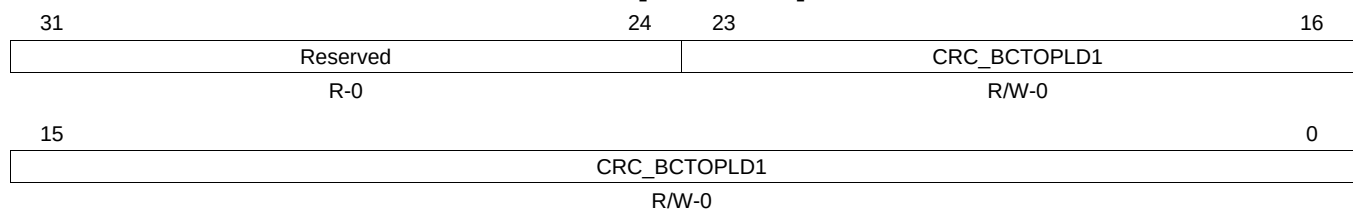
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-16. CRC Channel 1 Watchdog Timeout Preload Register A (CRC_WDTPLD1)
Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-0	CRC_WDTPLD1		Channel 1 Watchdog Timeout Counter Preload Register. This register contains the number of clock cycles within which the DMA must transfer the next block of data patterns. In Semi-CPU mode, this register is used to indicate the sector number for which the compression complete has last happened.

14.4.13 CRC Channel 1 Block Complete Timeout Preload Register B (CRC_BCTOPLD1)

Figure 14-21. CRC Channel 1 Block Complete Timeout Preload Register B (CRC_BCTOPLD1)
[offset = 50h]



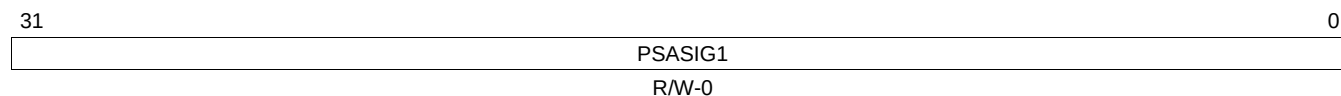
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-17. CRC Channel 1 Block Complete Timeout Preload Register B (CRC_BCTOPLD1)
Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-0	CRC_BCTOPLD1		Channel 1 Block Complete Timeout Counter Preload Register. This register contains the number of clock cycles within which the CRC for an entire block needs to complete before a timeout interrupt is generated.

14.4.14 Channel 1 PSA Signature Low Register (PSA_SIGREGL1)

Figure 14-22. Channel 1 PSA Signature Low Register (PSA_SIGREGL1) [offset = 60h]



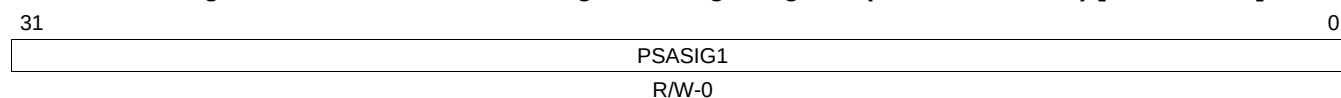
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-18. Channel 1 PSA Signature Low Register (PSA_SIGREGL1) Field Descriptions

Bit	Field	Description
31-0	PSASIG1	Channel 1 PSA Signature Low Register. This register contains the value stored at PSASIG1[31:0] register.

14.4.15 Channel 1 PSA Signature High Register (PSA_SIGREGH1)

Figure 14-23. Channel 1 PSA Signature High Register (PSA_SIGREGH1) [offset = 64h]



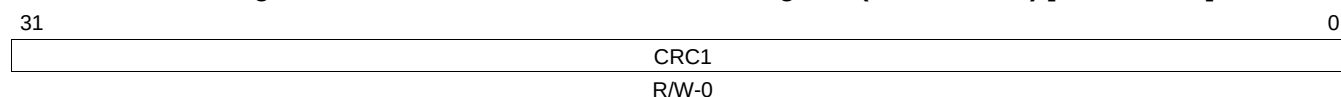
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-19. Channel 1 PSA Signature High Register (PSA_SIGREGH1) Field Descriptions

Bit	Field	Description
31-0	PSASIG1	Channel 1 PSA Signature High Register. This register contains the value stored at PSASIG1[63:32] register.

14.4.16 Channel 1 CRC Value Low Register (CRC_REGL1)

Figure 14-24. Channel 1 CRC Value Low Register (CRC_REGL1) [offset = 68h]



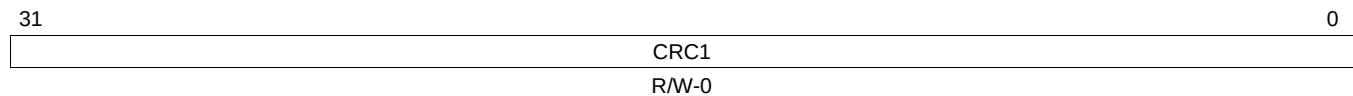
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-20. Channel 1 CRC Value Low Register (CRC_REGL1) Field Descriptions

Bit	Field	Description
31-0	CRC1	Channel 1 CRC Value Low Register. This register contains the current known good signature value stored at CRC1[31:0] register.

14.4.17 Channel 1 CRC Value High Register (CRC_REGH1)

Figure 14-25. Channel 1 CRC Value High Register (CRC_REGH1) [offset = 6Ch]



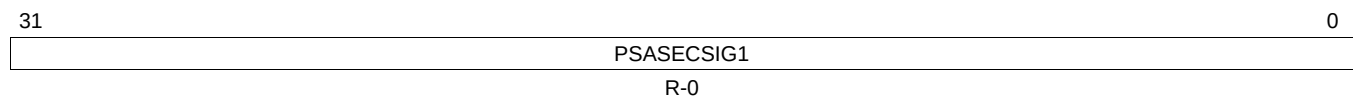
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-21. Channel 1 CRC Value High Register (CRC_REGH1) Field Descriptions

Bit	Field	Description
31-0	CRC1	Channel 1 CRC Value High Register. This register contains the current known good signature value stored at CRC1[63:32] register.

14.4.18 Channel 1 PSA Sector Signature Low Register (PSA_SECSIGREGL1)

Figure 14-26. Channel 1 PSA Sector Signature Low Register (PSA_SECSIGREGL1) [offset = 70h]



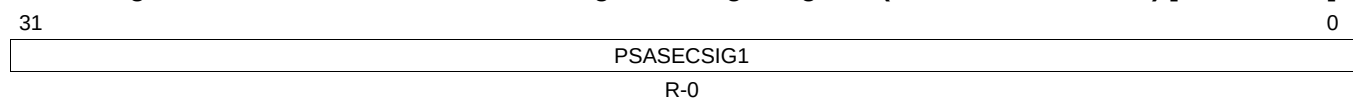
LEGEND: R = Read only; -n = value after reset

Table 14-22. Channel 1 PSA Sector Signature Low Register (PSA_SECSIGREGL1) Field Descriptions

Bit	Field	Description
31-0	PSASECSIG1	Channel 1 PSA Sector Signature Low Register. This register contains the value stored at PSASECSIG1[31:0] register.

14.4.19 Channel 1 PSA Sector Signature High Register (PSA_SECSIGREGH1)

Figure 14-27. Channel 1 PSA Sector Signature High Register (PSA_SECSIGREGH1) [offset = 74h]



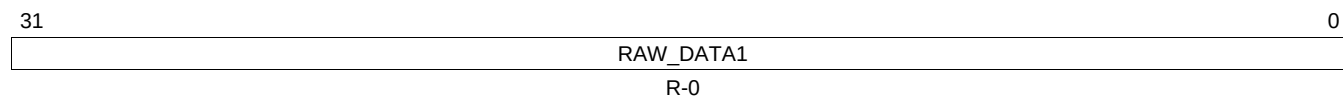
LEGEND: R = Read only; -n = value after reset

Table 14-23. Channel 1 PSA Sector Signature High Register (PSA_SECSIGREGH1) Field Descriptions

Bit	Field	Description
31-0	PSASECSIG1	Channel 1 PSA Sector Signature High Register. This register contains the value stored at PSASECSIG1[63:32] register.

14.4.20 Channel 1 Raw Data Low Register (RAW_DATAREGL1)

Figure 14-28. Channel 1 Raw Data Low Register (RAW_DATAREGL1) [offset = 78h]



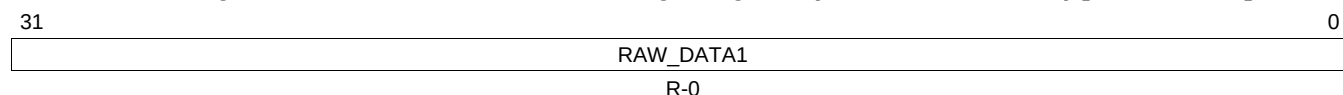
LEGEND: R = Read only; -n = value after reset

Table 14-24. Channel 1 Raw Data Low Register (RAW_DATAREGL1) Field Descriptions

Bit	Field	Description
31-0	RAW_DATA1	Channel 1 Raw Data Low Register. This register contains bits 31:0 of the uncompressed raw data.

14.4.21 Channel 1 Raw Data High Register (RAW_DATAREGH1)

Figure 14-29. Channel 1 Raw Data High Register (RAW_DATAREGH1) [offset = 7Ch]



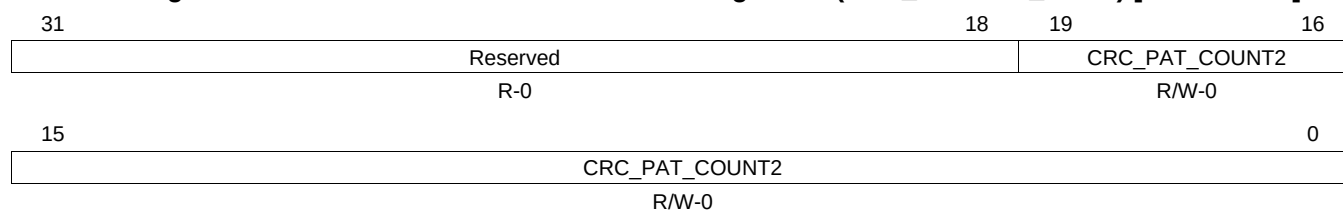
LEGEND: R = Read only; -n = value after reset

Table 14-25. Channel 1 Raw Data High Register (RAW_DATAREGH1) Field Descriptions

Bit	Field	Description
31-0	RAW_DATA1	Channel 1 Raw Data High Register. This register contains bits 63:32 of the uncompressed raw data.

14.4.22 CRC Pattern Counter Preload Register 2 (CRC_PCOUNT_REG2)

Figure 14-30. CRC Pattern Counter Preload Register 2 (CRC_PCOUNT_REG2) [offset = 80h]



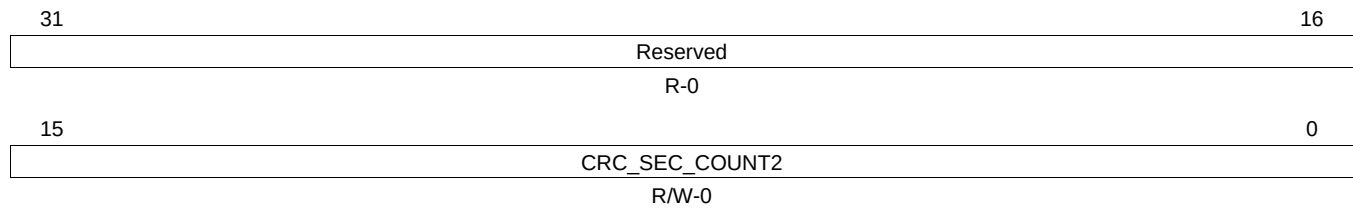
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-26. CRC Pattern Counter Preload Register 2 (CRC_PCOUNT_REG2) Field Descriptions

Bit	Field	Value	Description
31-20	Reserved	0	Reads return 0. Writes have no effect.
19-0	CRC_PAT_COUNT2		Channel 2 Pattern Counter Preload Register. This register contains the number of data patterns in one sector to be compressed before a CRC is performed.

14.4.23 CRC Sector Counter Preload Register 2 (CRC_SCOUNT_REG2)

Figure 14-31. CRC Sector Counter Preload Register 2 (CRC_SCOUNT_REG2) [offset = 84h]



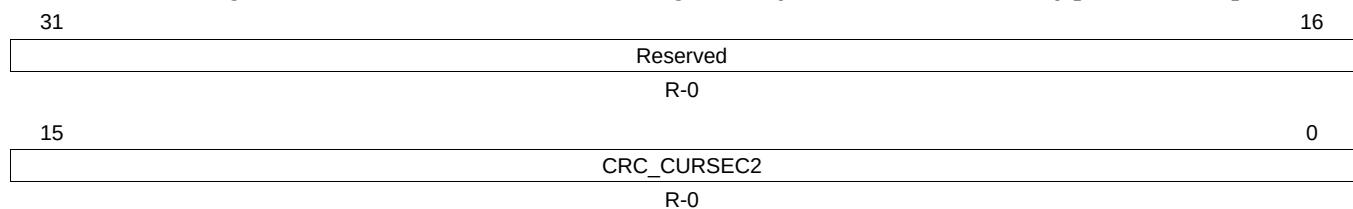
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-27. CRC Sector Counter Preload Register 2 (CRC_SCOUNT_REG2) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	CRC_SEC_COUNT2		Channel 2 Sector Counter Preload Register. This register contains the number of sectors in one block of memory.

14.4.24 CRC Current Sector Register 2 (CRC_CURSEC_REG2)

Figure 14-32. CRC Current Sector Register 2 (CRC_CURSEC_REG2) [offset = 88h]



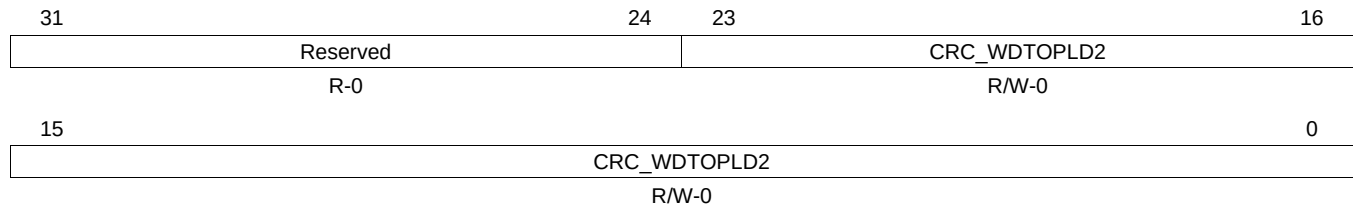
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-28. CRC Current Sector Register 2 (CRC_CURSEC_REG2) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	CRC_CURSEC2		Channel 2 Current Sector ID Register. In AUTO mode, this register contains the current sector number of which the signature verification fails. The sector counter is a free running up counter. When a sector fails, the erroneous sector number is logged into current sector ID register and the CRC fail interrupt is generated. The sector ID register is frozen until it is read and the CRC fail status bit is cleared by CPU. While it is frozen, it does not capture another erroneous sector number. When this condition happens, an overrun interrupt is generated instead. Once the register is read and the CRC fail interrupt flag is cleared it can capture new erroneous sector number. In Semi-CPU mode, this register is used to indicate the sector number for which the compression complete has last happened.

14.4.25 CRC Channel 2 Watchdog Timeout Preload Register A (CRC_WDTPLD2)

Figure 14-33. CRC Channel 2 Watchdog Timeout Preload Register A (CRC_WDTPLD2)
[offset = 8Ch]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-29. CRC Channel 2 Watchdog Timeout Preload Register A (CRC_WDTPLD2)
Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-0	CRC_WDTPLD2		Channel 2 Watchdog Timeout Counter Preload Register. This register contains the number of clock cycles within which the DMA must transfer the next block of data patterns. In Semi-CPU mode, this register is used to indicate the sector number for which the compression complete has last happened.

14.4.26 CRC Channel 2 Block Complete Timeout Preload Register B (CRC_BCTOPLD2)

Figure 14-34. CRC Channel 2 Block Complete Timeout Preload Register B (CRC_BCTOPLD2)
[offset = 90h]



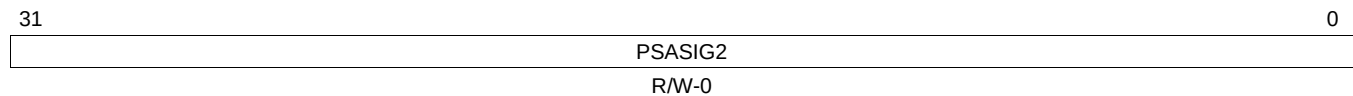
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-30. CRC Channel 2 Block Complete Timeout Preload Register B (CRC_BCTOPLD2)
Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-0	CRC_BCTOPLD2		Channel 2 Block Complete Timeout Counter Preload Register. This register contains the number of clock cycles within which the CRC for an entire block needs to complete before a timeout interrupt is generated.

14.4.27 Channel 2 PSA Signature Low Register (PSA_SIGREGL2)

Figure 14-35. Channel 2 PSA Signature Low Register (PSA_SIGREGL2) [offset = A0h]



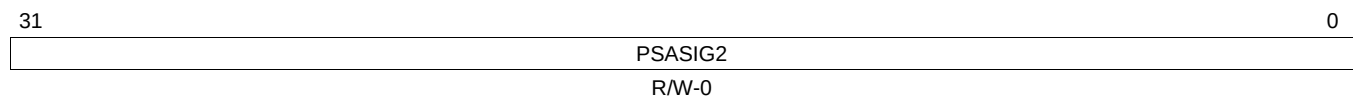
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-31. Channel 2 PSA Signature Low Register (PSA_SIGREGL2) Field Descriptions

Bit	Field	Description
31-0	PSASIG2	Channel 2 PSA Signature Low Register. This register contains the value stored at PSASIG2[31:0] register.

14.4.28 Channel 2 PSA Signature High Register (PSA_SIGREGH2)

Figure 14-36. Channel 2 PSA Signature High Register (PSA_SIGREGH2) [offset = A4h]



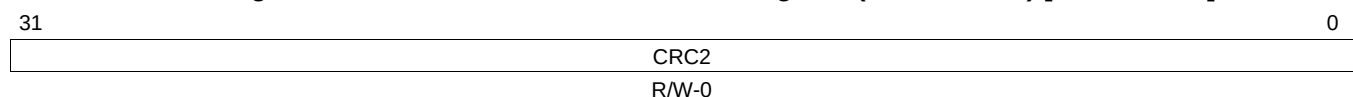
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-32. Channel 2 PSA Signature High Register (PSA_SIGREGH2) Field Descriptions

Bit	Field	Description
31-0	PSASIG2	Channel 2 PSA Signature High Register. This register contains the value stored at PSASIG2[63:32] register.

14.4.29 Channel 2 CRC Value Low Register (CRC_REGL2)

Figure 14-37. Channel 2 CRC Value Low Register (CRC_REGL2) [offset = A8h]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-33. Channel 2 CRC Value Low Register (CRC_REGL2) Field Descriptions

Bit	Field	Description
31-0	CRC2	Channel 2 CRC Value Low Register. This register contains the current known good signature value stored at CRC2[31:0] register.

14.4.30 Channel 2 CRC Value High Register (CRC_REGH2)

Figure 14-38. Channel 2 CRC Value High Register (CRC_REGH2) [offset = ACh]

31		0
CRC2		
R/W-0		

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-34. Channel 2 CRC Value High Register (CRC_REGH2) Field Descriptions

Bit	Field	Description
31-0	CRC2	Channel 2 CRC Value High Register. This register contains the current known good signature value stored at CRC2[63:32] register.

14.4.31 Channel 2 PSA Sector Signature Low Register (PSA_SECSIGREGL2)

Figure 14-39. Channel 2 PSA Sector Signature Low Register (PSA_SECSIGREGL2) [offset = B0h]

31		0
PSASECSIG2		
R-0		

LEGEND: R = Read only; -n = value after reset

Table 14-35. Channel 2 PSA Sector Signature Low Register (PSA_SECSIGREGL2) Field Descriptions

Bit	Field	Description
31-0	PSASECSIG2	Channel 2 PSA Sector Signature Low Register. This register contains the value stored at PSASECSIG2[31:0] register.

14.4.32 Channel 2 PSA Sector Signature High Register (PSA_SECSIGREGH2)

Figure 14-40. Channel 2 PSA Sector Signature High Register (PSA_SECSIGREGH2) [offset = B4h]

31		0
PSASECSIG2		
R-0		

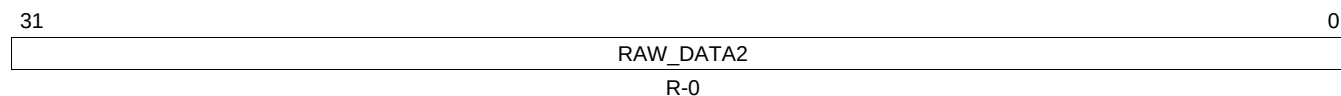
LEGEND: R = Read only; -n = value after reset

Table 14-36. Channel 2 PSA Sector Signature High Register (PSA_SECSIGREGH2) Field Descriptions

Bit	Field	Description
31-0	PSASECSIG2	Channel 2 PSA Sector Signature High Register. This register contains the value stored at PSASECSIG2[63:32] register.

14.4.33 Channel 2 Raw Data Low Register (RAW_DATAREGL2)

Figure 14-41. Channel 2 Raw Data Low Register (RAW_DATAREGL2) [offset = B8h]



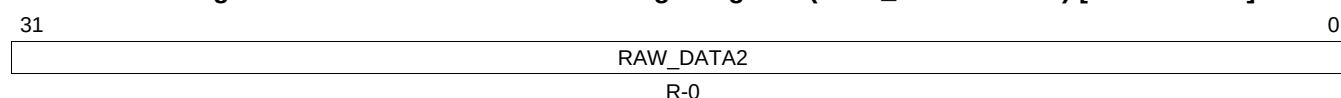
LEGEND: R = Read only; -n = value after reset

Table 14-37. Channel 2 Raw Data Low Register (RAW_DATAREGL2) Field Descriptions

Bit	Field	Description
31-0	RAW_DATA2	Channel 2 Raw Data Low Register. This register contains bits 31:0 of the uncompressed raw data..

14.4.34 Channel 2 Raw Data High Register (RAW_DATAREGH2)

Figure 14-42. Channel 2 Raw Data High Register (RAW_DATAREGH2) [offset = BCh]



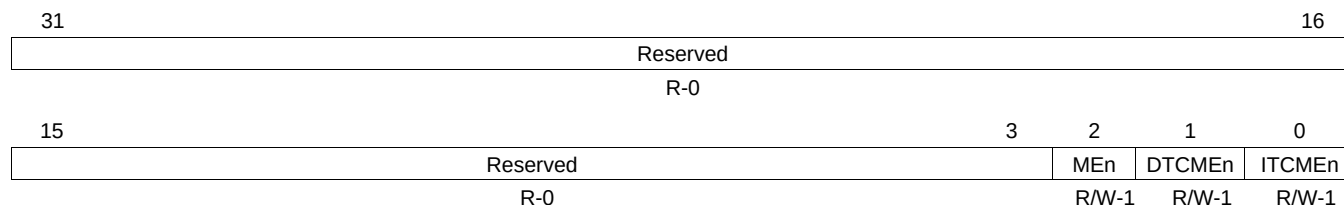
LEGEND: R = Read only; -n = value after reset

Table 14-38. Channel 2 Raw Data High Register (RAW_DATAREGH2) Field Descriptions

Bit	Field	Description
31-0	RAW_DATA2	Channel 2 Raw Data High Register. This register contains bits 63:32 of the uncompressed raw data..

14.4.35 Data Bus Selection Register (CRC_TRACE_BUS_SEL)

Figure 14-43. Data Bus Selection Register (CRC_TRACE_BUS_SEL) [offset = 140h]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 14-39. Data Bus Selection Register Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	MEn	0	Enable/disables the tracing of Peripheral Bus Master Tracing of Peripheral Bus Master has been disabled.
		1	Tracing of Peripheral Bus Master has been enabled.
1	DTCMEn	0	Enable/disables the tracing of data TCM Tracing of System Odd and Even RAM buses have been disabled.
		1	Tracing of System Odd and Even RAM buses have been enabled.
0	ITCMEn	0	Enable/disables the tracing of instruction TCM Tracing of Flash data bus has been disabled.
		1	Tracing of Flash data bus has been enabled.